

DAT540 Project Guidelines

The objective of group projects in this course is to solve problems by working with real-world datasets. The aim is to think like a data scientist and go through all the steps of a typical data science project. During this project, the student will learn about designing a research plan, preprocessing data, preparing input for the model, selecting an algorithm and training the model, interpreting the results, and evaluating and reporting the outcomes scientifically.

General Rules:

- Every project has a distinct dataset, and you must perform the exploratory analysis necessary for the final task.
 - Each group consists of 5 students.
 - Sign up for the group on the spreadsheet. A group leader is assigned within each group and is responsible for teamwork and coordination.
 - You can contact the project coordinator and meet him/her for refined consultation.
- For instance: (1) Outline of Data visualization, (2) Identifying the kind of ML technique to be applied, (3) Input feature(s), target variable, (4) Performance metrics.

General Assessment Criteria:

- Did the group complete all the steps of the data science project?
- Did the group formulate a valid research question?
- Did the group use the data to try to answer the research question? Is the presentation self-contained?
- Is the notebook submission fully explained and well-documented? Did the entire group contribute to the task.
- Does the work cite relevant sources, and are the authorities acknowledged correctly?
- Does the report and video clearly explain the various aspects of the project and justify the decisions made?

Key Notes for Video Submission

- Maximum Duration: 10 to 12 minutes
- Focus on demonstrating and explaining the code, not repeating the written report
- Show how the code works, what each major block does.
- Use the video to prove your understanding of the coding process and decisions

The table below outlines the guidelines for evaluation of your data science project in more details. It covers three delivery components and associated tasks: Code/Notebook, Written report, and Video presentation. The Data Exploration process carries high importance and should demonstrate strong understanding and insight.

Component	Task	Description	Evaluation Points
Notebook/ Code files	Data Exploration	Initial Data Review	Shape of dataset; data types; feature overview; missing values; duplicates; etc.
	Data Exploration	Cleaning	Handled nulls; removed duplicates; addressed inconsistencies; etc.
	Data Exploration	Visualization	At least 3 meaningful plots; insights clearly interpreted
	Data Exploration	Preprocessing	Type conversions; normalization or scaling; outlier handling when needed
	Data Exploration	Feature Selection	Identified key features; justification provided
	Model Selection	Model Justification	Explained suitability; aligned models with problem
	Model Selection	Model Variety	At least 3 models tested
	Model Selection	Research Question	Clearly defined problem or hypothesis
	Training	Data Preparation	Split data correctly; normalization applied if needed; nulls handled
	Training	Training Performance	Reported accuracy/error/metrics interpreted
	Validation & Evaluation	Evaluation Metrics	Metrics used for classification/regression/clustering (for example: accuracy, regression error, Silhouette Coefficient)
	Validation & Evaluation	Parameter Tuning	Evidence of tuning or optimization if needed
	Validation & Evaluation	Predictions	Tested on unseen data; showed sample outputs
Report		Structure & Clarity	Clear sections; concise writing; max 4 pages
Report		Problem Definition	Dataset described; objective clearly stated
Report		Method Summary	Key preprocessing and models summarized
Report		Results & Insights	Performance summarized; insights explained
Report		Conclusion	Findings; challenges; future improvements
Video		Code Walkthrough	Explained core parts of the code; demonstrated how each part works
Video		Implementation Decisions	justified coding choices (functions, libraries, preprocessing steps) and why they were implemented that way
Video		Model Coding & Training	Showed how models were implemented in code, including training loops and parameter settings
Video		Interpreting Outputs	Explained evaluation results directly from code execution (plots, predictions)